1. Prove that the following two statements are equivalent (that is, prove that (i) $\iff$ (ii)):

   (i) If $a, b \in \mathbb{R}$ and $a < b$, then there exists $r \in \mathbb{Q}$ such that $a < r < b$.

   (ii) Given $x \in \mathbb{R}$ and $\varepsilon > 0$, there exists $r \in \mathbb{Q}$ such that $r \in (x - \varepsilon, x + \varepsilon)$.

2. Use the Bisection Method to finish the following proof of statement (ii). By Question 1, this is an alternative proof that there is a rational between every pair of real numbers.

   *Proof.* Let $x \in \mathbb{R}$ and $\varepsilon > 0$. If $x$ is rational, then $x \in (x-\varepsilon, x+\varepsilon)$, so statement (ii) is true. Now suppose $x$ is irrational. Since $\mathbb{N}$ is not bounded above, there exists a positive integer greater than $x$. Let $b_1$ be the smallest positive integer greater than $x$, and let $a_1 = b_1 - 1$. Since $x$ is not an integer, $a_1 < x < b_1$; that is, $x$ is in the interval $I_1 := [a_1, b_1]$. Now, either $x \in [a_1, \frac{a_1+b_1}{2}]$ or $x \in [\frac{a_1+b_1}{2}, b_1]$. If $x = \frac{a_1+b_1}{2}$, we are done. Otherwise let $I_2 := [a_2, b_2]$ be the subinterval $[a_1, \frac{a_1+b_1}{2}]$ or $[\frac{a_1+b_1}{2}, b_1]$ of $I_1$ which contains $x$. Note that the endpoints of $I_2$ are rational. **(Complete the proof below.)**

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